Telegram Ads Forecasting – Model v1 Report

Karim Khabib

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1 Executive Summary

The goal is to predict ad views (`VIEWS`) for Telegram channels given only three input features available at inference time: `CPM`, `CHANNEL_NAME`, and `DATE`. The final Model v1 is an explainable CatBoost baseline that combines channel-level statistics, CPM-based effects, and yearless seasonality.

Public leaderboard score (best run): **0.927**. The main limitation is label noise and a weak global CPM–VIEWS relationship. Large gains likely require external channel metadata (subscribers, ER, topic).

2 Problem Statement

We must predict `VIEWS` for each row in the test set. The training set (`AllData.csv`) has 7 columns, while the test set includes only 3 input features. Therefore, the model must be built using only the features available at inference time.

3 Data Overview

3.1 Train/Test Summary

- Train size: 142,609 rows, 7 columns.
- Test size: 318,722 rows, 4 columns (`VIEWS` missing).
- Missing values: none in key fields.
- Date range (train): 2024-10-02 to 2025-12-17 (441 days).
- Unique channels (train): 35,957.
- Unique channels (test): 6,751 (all appear in train).

3.2 Distributions and Heavy Tails

Both `VIEWS` and `CPM` have heavy tails. This motivates training on log-transformed targets and features.

3.3 Channel Sparsity

Channels are highly sparse: median rows per channel = 2, 90% have at most 9 rows, and the maximum is 137. This requires shrinkage toward global statistics to avoid overfitting.

4 EDA Findings

4.1 Global vs. Within-Channel CPM Relationship

Global correlation between `CPM` and `VIEWS` is weak. However, within-channel correlations are often positive. This suggests that CPM is informative only after conditioning on channel.

4.2 Seasonality

Yearless seasonality (day-of-week and day-of-year cycles) shows mild patterns, so we keep simple date features without absolute time trends.

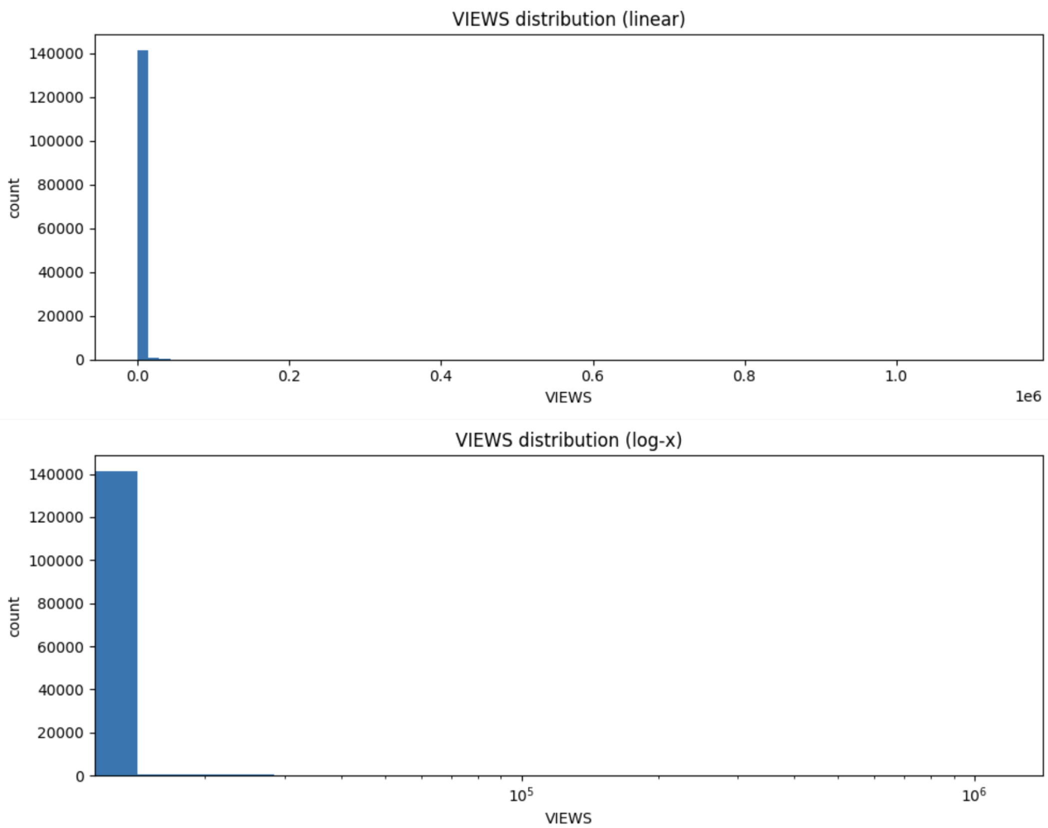

Figure 1: Distribution of VIEWS (linear and log scales).

5 Modeling Approach

5.1 Target Transformation

We predict $\log(1 + \texttt{VIEWS})$ and invert predictions with $\exp(x) - 1$. This stabilizes variance and reduces the impact of extreme outliers.

5.2 Features

- **CPM features:** `log_cpm`, `cpm_to_ch_median`.
- **Channel features:** smoothed median of log views, smoothed CTR, smoothed actions rate, CPM slope, and expected log prediction `ch_log_pred`.
- **Date features:** `dow`, `is_weekend`, `month`, `doy_sin`, `doy_cos`.

All channel-level statistics are computed on train only and smoothed with shrinkage factors to avoid overfitting.

5.3 Model and Post-Processing

- Model: `CatBoostRegressor` with MAE objective.

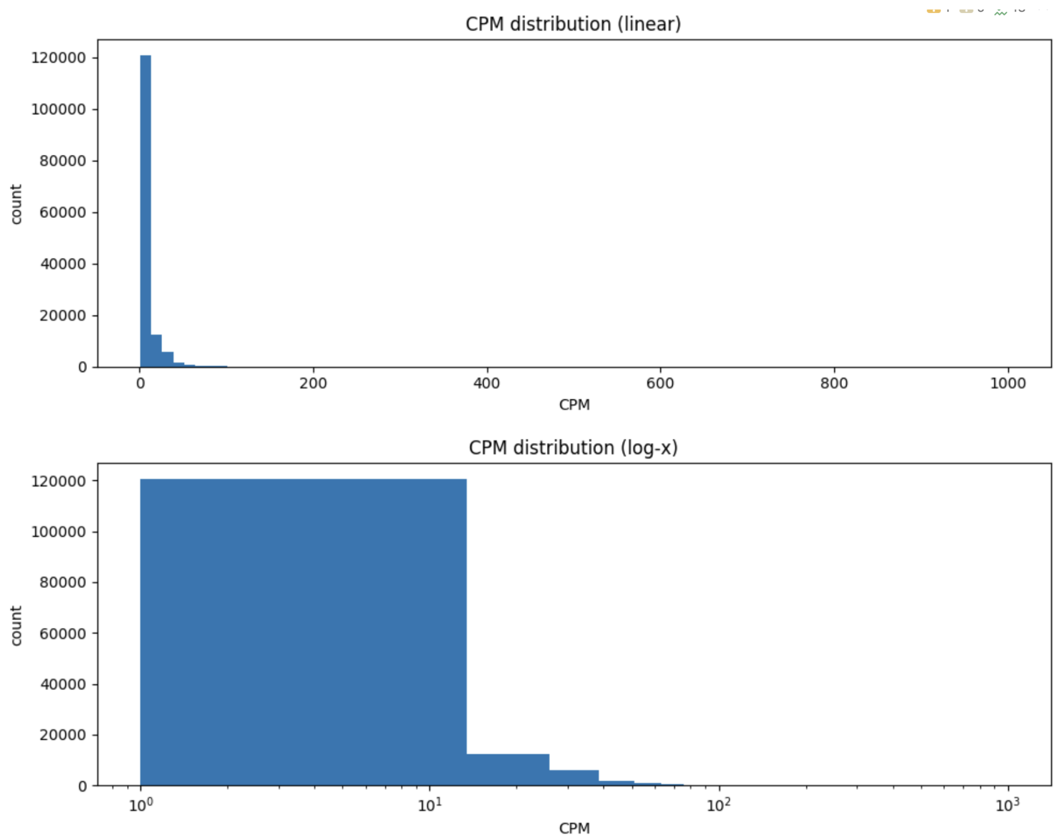

Figure 2: Distribution of CPM (linear and log scales).

- Post-processing: prediction blending with channel baselines, optional clipping by per-channel quantiles, optional scaling.

6 Training and Validation

We use a time-based holdout of the last 30 days for quick local checks. However, the public test set contains earlier dates, so local validation is not fully aligned with the leaderboard.

7 Metric Exploration

The public metric is not documented. We probed it with constant predictions:

Constant prediction	Public score
0	1.0000
300	0.9427
1000	1.0938
3000	1.9048

This behavior suggests a relative-error metric (MAPE/SMAPE-like), where large overestimates are heavily penalized. Therefore, conservative predictions and clipping help.

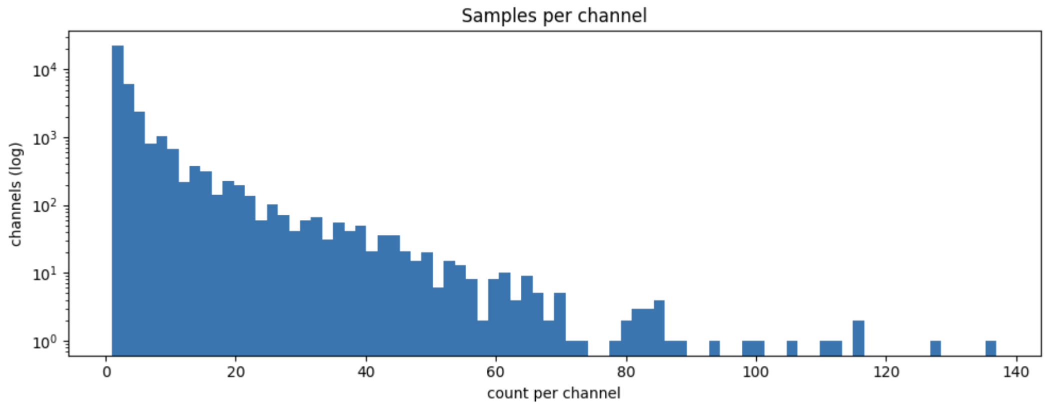

Figure 3: Histogram of rows per channel.

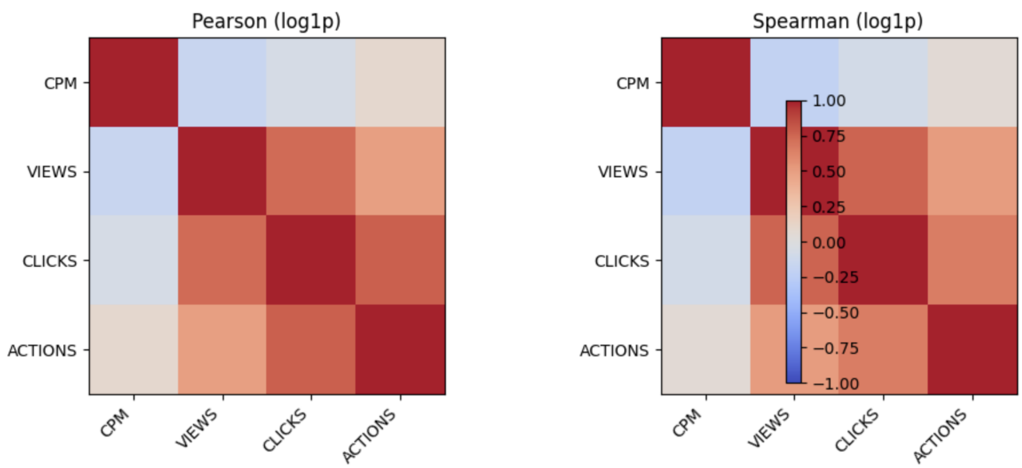

Figure 4: Pearson/Spearman correlation on log-transformed variables.

8 Results

Best public score achieved with Model v1: **0.927**. This is stronger than naive baselines but still far from ideal, primarily due to noise and missing channel metadata.

9 Error Analysis

- The same (`CPM`, `CHANNEL_NAME`, `DATE`) can map to different `VIEWS`, indicating label noise.
- Most channels have very few examples, so channel statistics are noisy.
- CPM is weak globally; it matters mostly within channels.

10 Limitations

- Only three features are available at inference time.
- Channel metadata (subscribers, ER, topic) is missing.

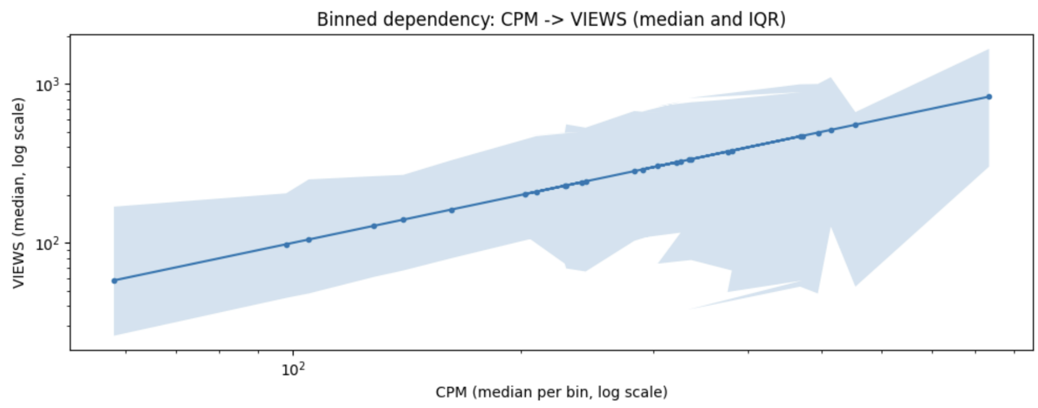

Figure 5: Binned CPM – VIEWS dependency (median and IQR).

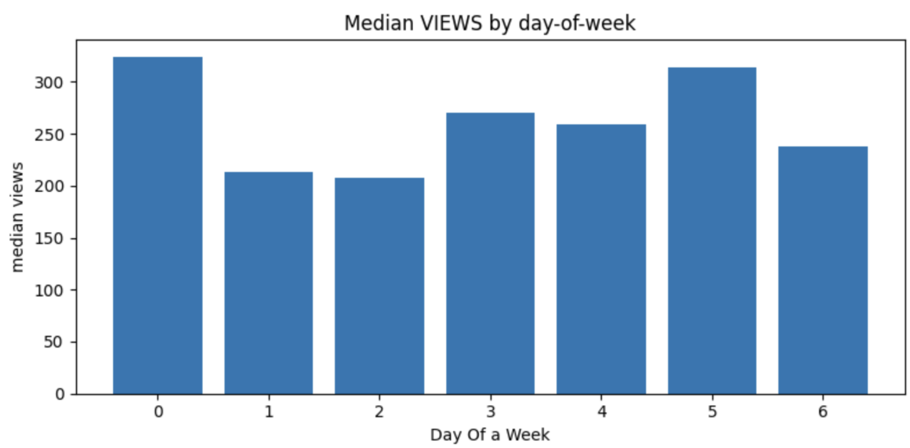

Figure 6: Average VIEWS by day of week.

- Validation is misaligned with the test temporal distribution.
- The evaluation metric is unknown and favors conservative predictions.

11 Future Work

- Enrich channels with external metadata (subscribers, ER, topic).
- Try monotonic constraints or isotonic regression for CPM within channels.
- Use per-channel KNN over CPM as a local smoother.
- Build a better validation scheme aligned with test distribution.